

- [Just Gimme What Works Advanced Stuff](#)

The SNPs That We Were Looking At

RS#	Call	Variant Allele	Gene	Variation	Result
rs1051266	TC	T	SLC19a1		+/-
rs2236225	AG	A	MTHFD1	G1958A	+/-
rs1801131	TT	G	MTHFR	A1298C	-/-
rs1801133	AA	A	MTHFR	C677T	+/+
rs7946	TC	T	PEMT	5465G>A	+/-

-/- = not present; +/- = heterozygous genotype; +/+ = homozygous genotype; +/-* = hemizygous genotype (male X chromosome).

Your Genetic Variants and Your Methylfolate Score

The scores below estimate the predicted decrease in activity, given your genotype, associated with the folate transporter (SLC19a1), the enzyme that converts tetrahydrofolate to 5,10-methylenetetrahydrofolate (MTHFD1), and the enzyme that converts 5,10-methylenetetrahydrofolate to 5-methyltetrahydrofolate (MTHFR).

SLC19A1 Score: 25% decrease

MTHFD1 Score: 13% decrease

MTHFR Score: 75% decrease

We then multiply these decreases together to yield a “methylfolate score” that estimates the combined decrease in methylfolate production:

Your Methylfolate Score: 84% decrease

Your Choline Requirement and Your Methylfolate Score

Based on studies in men and women homozygous for MTHFR C677T, a 75% loss in methylfolate production is taken to indicate a doubling of the choline requirement. The choline calculator takes the degree to which your methylfolate score approaches or exceeds this threshold and adjusts your choline requirement accordingly. The adequate intake (AI) assigned to adult men of 550 mg/d is assumed as the basal requirement, and it is not adjusted for sex due to lack of clear evidence supporting decreased needs for women. An “egg yolk equivalent” is 136 mg of choline, the amount in one large 20-gram egg.

PEMT and Bad Consequences

The comment about possible bad consequences of failing to meet your choline requirement is based on the references listed under the heading “References related to PEMT” below.

References

Single nucleotide polymorphisms in the human reduced folate carrier: characterization of a high-frequency G/A variant at position 80 and transport properties of the His(27) and Arg(27) carriers. [PMID: 11705857]

The MTHFD1 p.Arg653Gln variant alters enzyme function and increases risk for congenital heart defects. [PMID: 18767138]

A candidate genetic risk factor for vascular disease: a common mutation in methylenetetrahydrofolate reductase. [PMID: 7647779]

A second common mutation in the methylenetetrahydrofolate reductase gene: an additional risk factor for neural-tube defects? [PMID: 9545395]

Genetic impairments in folate enzymes increase dependence on dietary choline for phosphatidylcholine production at the expense of betaine synthesis. [PMID 27342765]

Choline intake exceeding current dietary recommendations preserves markers of cellular methylation in a genetic subgroup of folate-compromised men. [PMID 20220206]

References related to PEMT

Phosphatidylethanolamine N-methyltransferase gene rs7946 polymorphism plays a role in risk of nonalcoholic fatty liver disease: evidence from meta-analysis. [PMID 26636496]

Common genetic polymorphisms affect the human requirement for the nutrient choline. [PMID 16816108]

Genetic Variation in Choline-Metabolizing Enzymes Alters Choline Metabolism in Young Women Consuming Choline Intakes Meeting Current Recommendations. [PMID 28134761]

More Tools

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